What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

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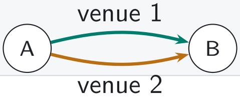
How we reconstruct a route from one transaction

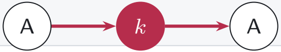
Direct swap
one transaction

Sequential vehicle route
one transaction

Parallel split
one transaction



Round trip
one transaction

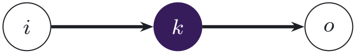


Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.

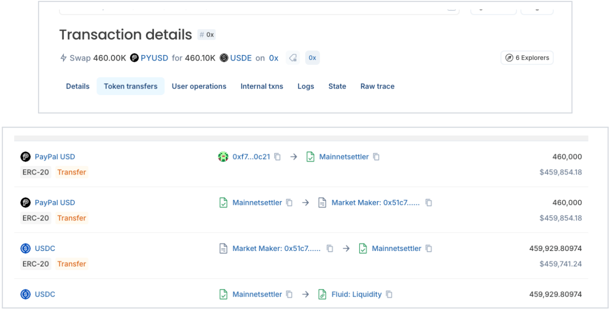


Count-weighted dominance
Share of indirect routes that pass through k

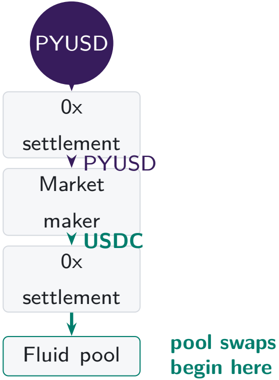
Value-weighted dominance
Share of comparable routed value that passes through k

The transaction starts with PYUSD

EXPLORER RECORD



SETTLEMENT HANDOFF

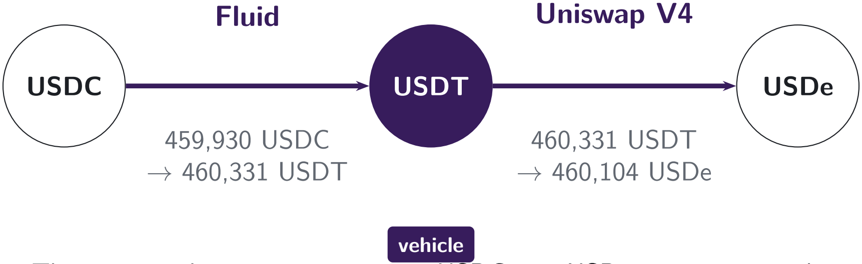


PYUSD reaches 0x settlement; the market maker supplies USDC for execution in the pools.

Note: The explorer screenshot and transfer trace refer to transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

The pool route continues through USDT

Full transaction path: PYUSD settlement $\longrightarrow$ **USDC** $\longrightarrow$ **USDT** $\longrightarrow$ **USDe**

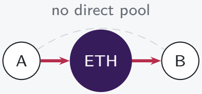


The connected component converts USDC into USDe across two pools.
USDT is the vehicle because it links the two legs.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Architecture changes routing opportunities

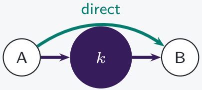
V1 mandatory hub



Only ETH–token pools: token-to-token execution must pass through ETH.

Hub use is imposed

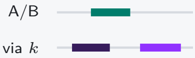
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

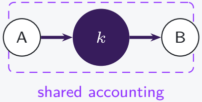
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



Two pools still execute; intermediate token movement may be netted.

Settlement can diverge from route use

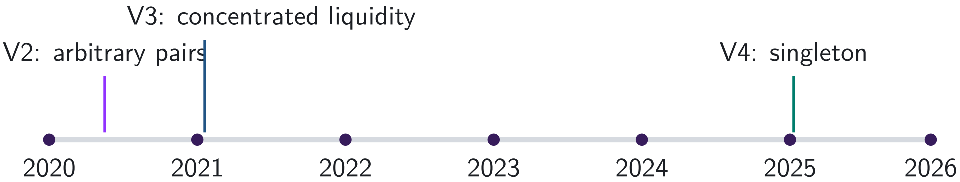
Architecture feasible rules

Exposure realised pool and route use

Calendar date timing only

The sample spans Uniswap V1 through V4

DATA



Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

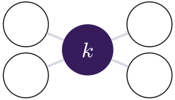
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

Excess vehicle use

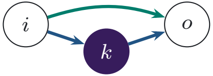
$X_k = D_k \div E_k$
E_k: endpoint share of k
$X_k > 1$: intermediary use exceeds the endpoint benchmark

Separate characteristics of the asset and route set



Network position

position of k in the feasible graph

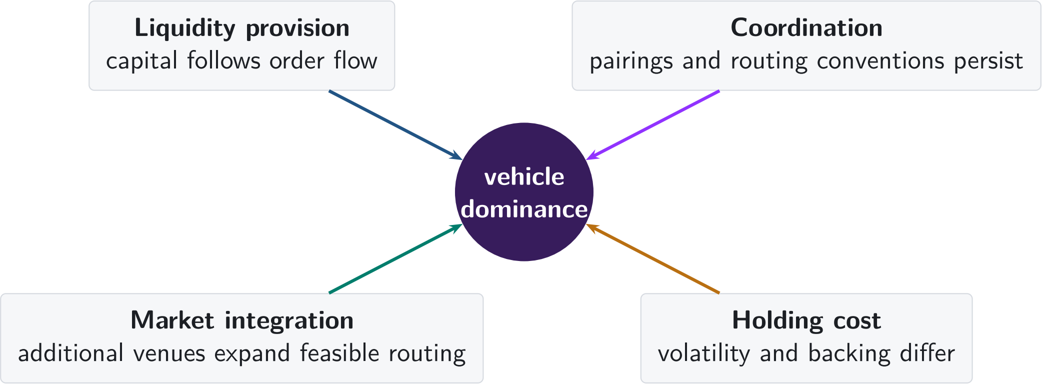


Execution cost

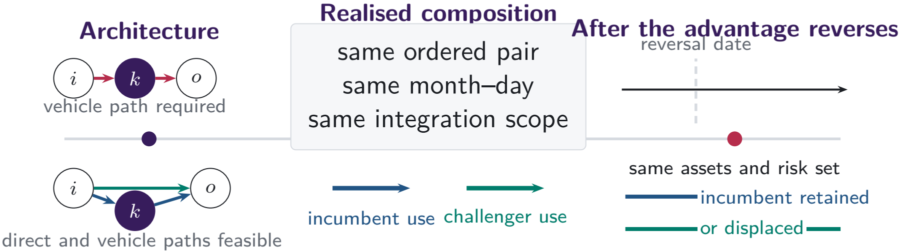
which route returns more at the same state

Endpoint demand supplies the benchmark; network position and same-state route costs remain separate characteristics.

Why might one asset become the dominant vehicle?

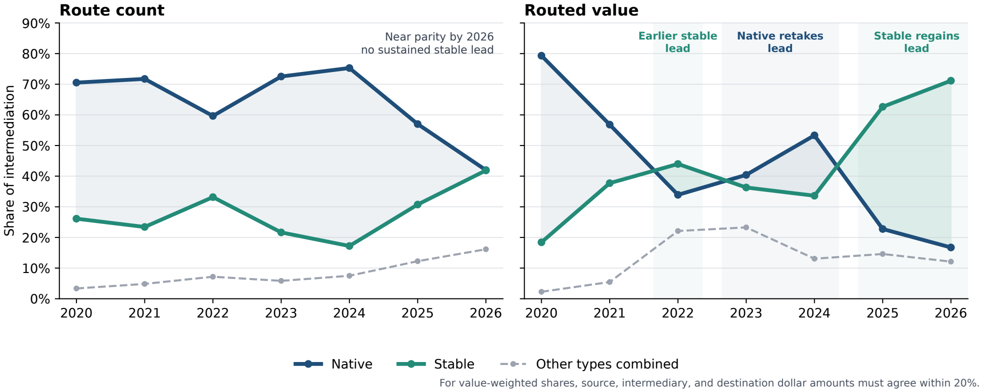


Architecture changes the set of feasible routes



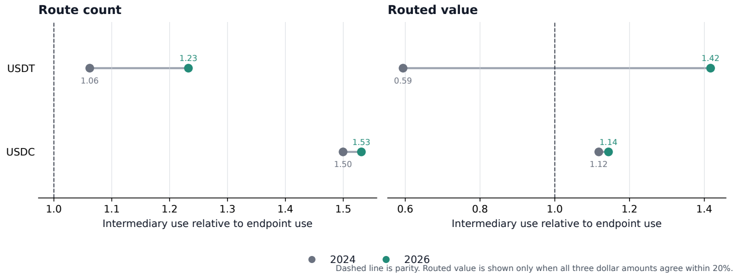
Conceptual design sequence — no estimates shown

Stablecoins regain the lead in routed value



Note: Annual shares of observed vehicle routes. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets.

USDT drives the rise in stablecoin intermediation



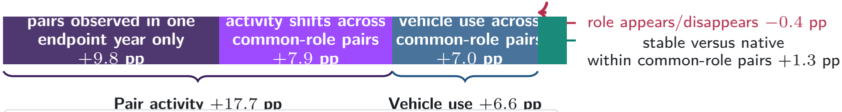
USDT crosses parity by routed value and extends its count advantage; USDC changes little.

93.7% of the count-share change and 85.6% of the routed-value change occur within the single- and cross-venue categories.

Note: Excess use is intermediary share divided by endpoint share on the same route perimeter. The comparison uses the same January–June dates in 2024 and 2026.

Pair activity and vehicle use: 94.9% of the gain

Stable share: 16.9% $\longrightarrow$ 42.5% $+25.7$ **pp**

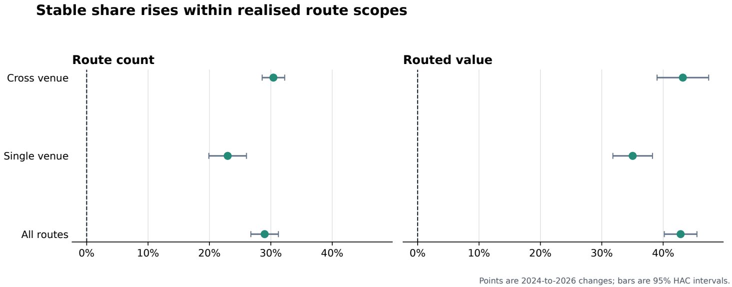


Among pairs observed in both endpoint years, stable share still rises
26.1% $\longrightarrow$ 41.9% $+15.8$ **pp**

Dollar-weighted routes show the same broad allocation. Stable share rises $+42.8$ pp, with $+26.2$ pp from activity weights among continuing pairs and only 0.0 pp from stable-for-native substitution within them.

Note: The sample contains exact two-leg routes observed in 2024 and January–June 2026. Matched estimates compare the same ordered endpoint pair, calendar date, and single- versus cross-venue route category. Dollar-weighted calculations retain routes for which the source, intermediary, and destination dollar amounts agree within 20%.

Stablecoin use rises within each venue scope



The increase appears within both realised route scopes, so a change in the single- versus cross-venue mix cannot account for the aggregate rotation. Cross-venue status remains selected; the comparison does not identify an effect of integration, routing software, or protocol architecture.

Note: Changes compare the same January–June dates in 2024 and 2026. Bars show 95% calendar-HAC confidence intervals.

Does liquidity move before vehicle use or follow it?

LIQUIDITY PRECEDES USE

More capital or provider inflow in vehicle-linked pools at t

predicts

More routes through the same vehicle at $t + \tau$

Liquidity supply may help a vehicle attract later routes.

USE PRECEDES LIQUIDITY

More routes through a vehicle at t

predicts

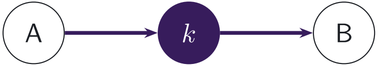
More capital or provider inflow in its pools at $t + \tau$

Providers may instead follow routed demand.

V2 capital stocks and V3 provider flows are tested separately at fixed lags; either ordering may reflect common demand or routing shocks.

V4 separates the economic route from physical settlement

ECONOMIC ROUTE



Pool calls route
through intermediary k.

Prediction: fewer token transfers
for comparable two-pool routes.

MATCHED COMPARISON

- Compare routes in the same week, endpoint pair, intermediary, direction, and trade-size group.
- Record route use and token transfers separately.
- Compare feasible alternatives before interpreting vehicle choice.

Challenger cost advantage predicts subsequent vehicle share

30-day outcome	$\hat{\beta}$	SE	t
Change in challenger vehicle share	4.825	0.366	13.17
Change in incumbent vehicle share	-3.872	0.420	-9.22
Challenger leads after 30 days	9.947	0.486	20.48

Note: $\hat{\beta}$ is the coefficient on the challenger's cost advantage. Outcomes are measured 30 days later. The sample contains $N = 57,989$ endpoint-pair–intermediary observations and 2,213 date clusters; standard errors are clustered by date and use the CR1 rank correction.

The estimates are large in the small set of endpoint markets and intermediary assets we can follow. They remain descriptive because the panel does not compare each market against a fixed set of feasible routes.

Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every token pool pairs with ETH, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH pool in 2026

97.9%

of pairs first traded in 2026 include WETH

V2 converted ETH pairing from a protocol rule into a market choice. Native-asset concentration remained high through pair formation and realised trading.

Note: V1 counts linked token-to-ETH and ETH-to-token swaps with matching ETH quantities in one transaction. V2 shares use single-leg Uniswap V2 trades and newly traded pairs; they describe that venue rather than market-wide vehicle dominance.

Stablecoin dominance grew through changes in routed markets

- Stablecoins lead routed value in 2022 and again in 2025–2026, but fall behind native assets in 2023–2024.
- USDT drives the post-2024 rise in excess stablecoin intermediation; USDC changes little.
- Changes in pair coverage, trading across pairs, and indirect-route use account for $+24.4$ pp of the $+25.7$ pp aggregate increase. Stablecoin share within continuing vehicle-using pairs adds $+1.3$ pp.

Stablecoins became dominant in routed value as new endpoint markets formed and activity shifted across continuing markets.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

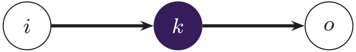
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

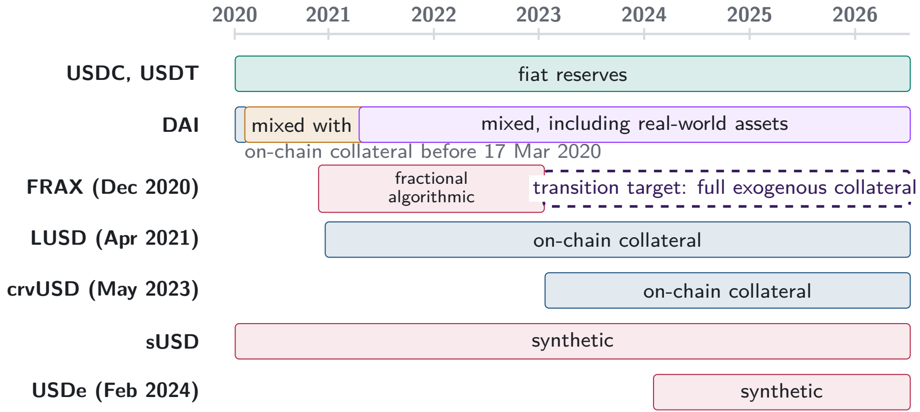
A1. One reconstructed route is one unit



one route unit
two swap legs

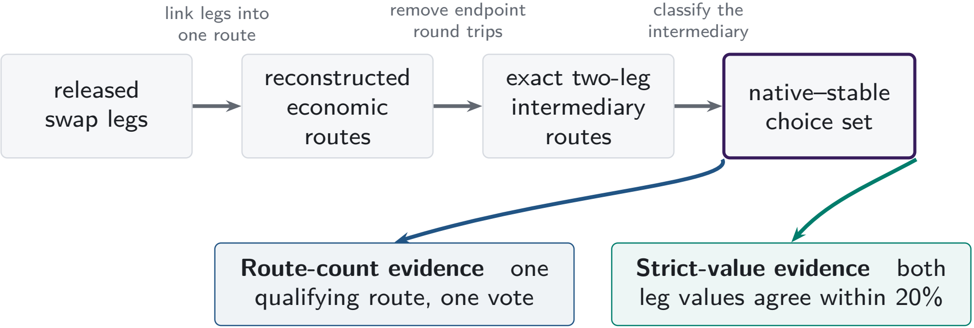
- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Stablecoin backing cannot be treated as fixed



Note: Bars begin at public launch or the sample start for older tokens. DAI's segments mark the backing regimes admitted by the classification; the dashed FRAX arrow is a governance target, not an observed collateral share.

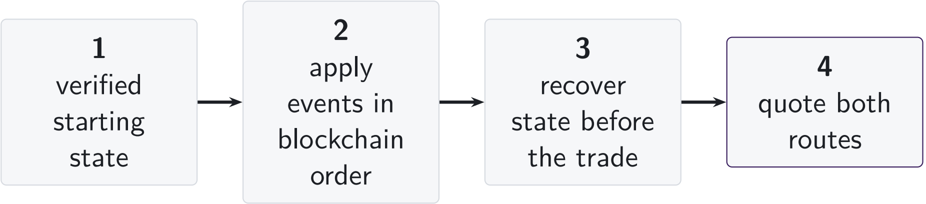
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

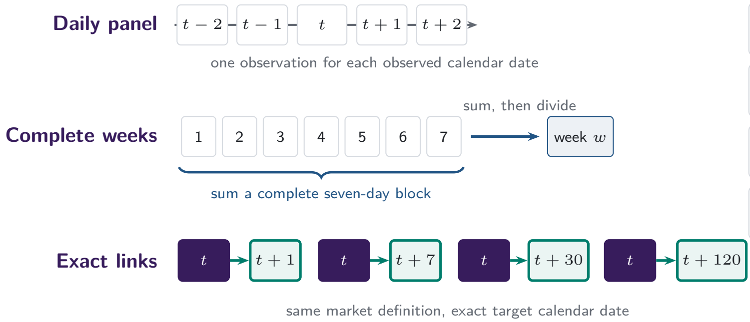
■ required state
□ hooks outside comparison

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.

same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

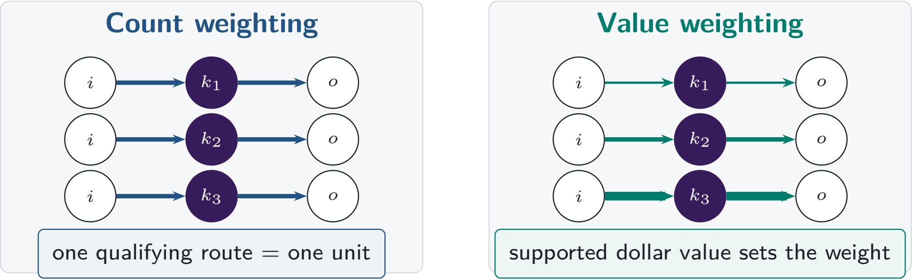
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_j n^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

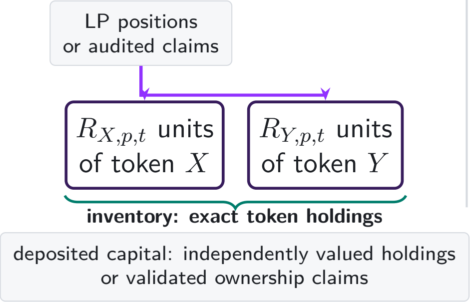
$$E^N_{k,t} = \frac{n^E_{k,t}}{\sum_j n^E_{j,t}}$$

How often asset k is what traders buy or sell.

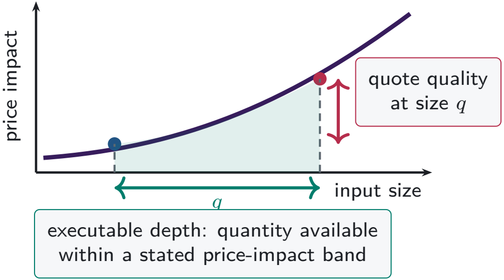
Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are different objects

Pool balance sheet at time t

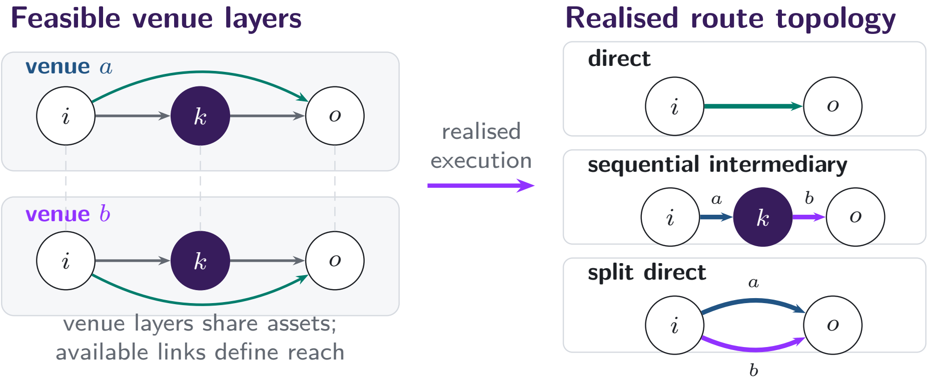


Execution from that state



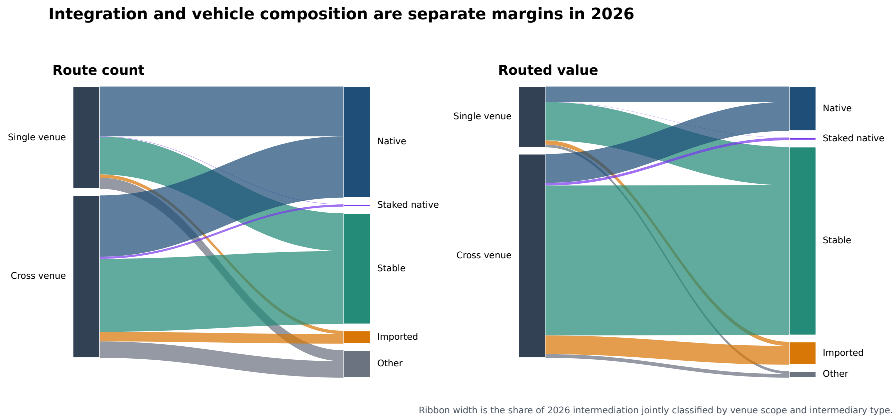
Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage